

UNIT 8: ECOLOGICAL CONCEPTS

8.1 – BEHAVIORAL ECOLOGY

Behavioral Ecology Outline

- Inheritance Influences Behavior
- The Environment Influences Behavior
- Animal Communications
- Behaviors that Increase Fitness

Inheritance Influences Behavior

- **Behavior** - any action that can be observed and described
- Nature (inherited) versus nurture (environmental) questions are still debated
 - ▣ Today, most evidence points to the conclusion that most behaviors have a genetic basis
 - ▣ May be tied to nervous and endocrine systems

Inheritance Influences Behavior

- Experiments That Suggest that Behavior has a Genetic Basis
 - ▣ Nest-Building Behavior in Lovebirds
 - Fischer lovebirds and peach-faced lovebirds build nests and carry leaves in different ways (Peach cut shorter strips and carry in their feathers to nest)
 - Hybrid offspring between these two species had difficulty carrying nesting materials
 - ▣ Food Choice in Garter Snakes
 - Inland populations are aquatic and feed on frogs and fish while coastal populations are terrestrial and feed primarily on slugs
 - Inland populations refuse to consume slugs in the laboratory, however, hybrid offspring between these two species accept slugs as a food source
 - ▣ Twin Studies in Humans
 - Human twins separated at birth and raised under different environmental conditions have been shown to have similar food preferences, activity patterns, and select mates with similar characteristics
 - Scott & Mark Kelly
 - ▣ Nurturing Behavior in mice
 - Maternal behavior post-birth found to be dependent on the presence of a single gene termed *fosB*

The Environment Influences Behavior

□ **Fixed Action Patterns (FAP's)**

- ▣ Unchanging behavioral response
- ▣ Increasingly, scientists have found behaviors that were thought to be FAP's could improve with practice

□ **Instinct & Learning**

- ▣ A durable change in behavior brought about by experience
 - Ex: pecking behavior in laughing gulls (pecking leads to food)

□ **Imprinting**

- ▣ Imitation of behavior observed during sensitive period (period of time when a particular behavior develops)
 - Ex: Birds follow any moving object after birth [Vid1](#)

The Environment Influences Behavior

□ **Social Interactions and Learning**

- ▣ Avian brain is especially sensitive to acoustical stimuli during a sensitive period
 - Social experience appears to have an even stronger influence over development of singing
 - Ex: white-crowned sparrow signing

□ **Associative Learning**

- ▣ Any change in behavior that involves an association between two events
 - **Classical conditioning**
 - The paired presentation of two different types of stimuli causes an animal to form an association between them
 - Ex: Pavlov - dogs salivate when presented with food. [Vid1](#) [Vid2](#)
 - **Operant Conditioning**
 - Gradual strengthening of stimulus-response connections
 - Trick-training in animals [Vid 1](#)

The Environment Influences Behavior

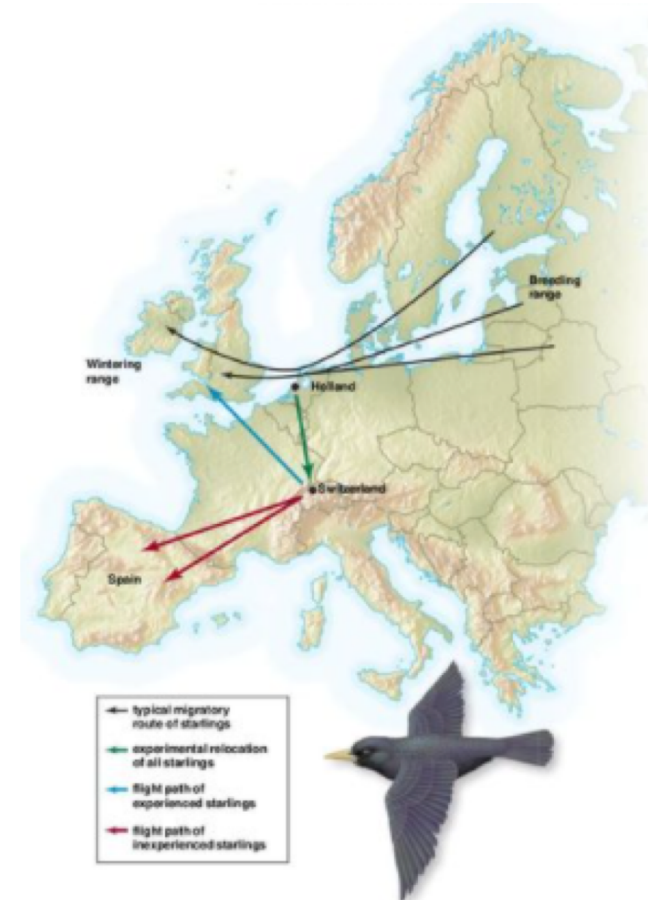
□ Orientation and Migratory Behavior

- **Orientation** - The ability to travel in a particular direction
 - Many birds use the sun or star for cues
- **Migration** - Long-distance travel from one location to another
 - Ex: Starling migration
- **Navigation** - The ability to change direction in response to environmental cues
 - Clues may come from the Earth's magnetic field

□ Cognitive Learning

- Learning through observation, imitation, and insight
 - **Insight learning** - Solving a problem without prior experience
 - Animal appears to draw upon prior experience with other circumstances to solve the problem.

[Vid1](#)



Animal Communications

- Some animals are largely solitary and join with a member of the opposite sex only to reproduce.
- Others pair, bond, and cooperate in raising offspring.
- **Societies** - Members are organized in a cooperative manner extending beyond sexual or parental behavior
- **Communication Behavior**
 - ▣ Communication is an action by a sender that influences the behavior of a receiver
 - ▣ May be purposeful, but does not have to be

Animal Communication

□ Chemical Communication

- ▣ **Pheromones** are chemical signals that are passed between members of the same species
 - Some animals are capable of secreting different pheromones, each with a different meaning

□ Auditory communication

- ▣ Faster than chemical communication
- ▣ Effective both day and night
- ▣ Can be modified by loudness, pattern, repetition, and duration
- ▣ Language is the ultimate auditory communication

□ Visual communication

- ▣ Allows animals to signal others without chemical or auditory messages
- ▣ Visual signals are most often used
 - By species that are active during the day
 - In contests between males who make use of threat postures
 - To establish dominance



Animal Communication

□ **Tactile Communication**

▣ Occurs when one animal touches another

- Gull chicks peck at the parent's beak in order to induce the parent to feed them
- Foraging honeybees use tactile communication to impart information about the environment
 - Honeybees return to the hive and perform a waggle dance
 - The dance indicates the distance and direction of a food source
 - [Vid 1](#)

Behaviors that Increase Fitness

□ Behavioral Ecology

- Assumes that behavior is subject to natural selection
 - Behavior has a genetic basis
 - Some behaviors lead to increased survival and number of offspring
- Behaviors of animals we observe today has adaptive value
- Territoriality, Reproductive Strategies, Social Behavior, Altruistic Behavior

Behaviors that Increase Fitness

□ Territoriality and Fitness

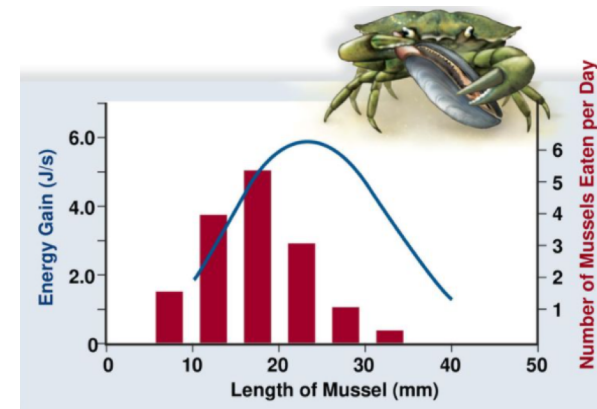
▣ **Territoriality** is protecting an area against other individuals

- Male gibbons maintain their territory by singing and fight to defend their territory
- Defending a territory costs energy
- Benefits of territoriality include a source of food, the right to one or more females, a place to rear young, and a place providing protection from predators

▣ Territoriality is more likely to occur during times of reproduction

▣ **Foraging for Food**

- Animals must acquire a food source that will provide more energy than the effort of acquiring the food
- The **optimal foraging model** states that it is adaptive for foraging behavior to be as energetically efficient as possible



Behaviors that Increase Fitness

□ Reproductive Strategies and Fitness

▣ Polygamous

- Males mate with multiple females
- Females invest more in the offspring

▣ Polyandrous

- One female mates with more than one male
- The environment cannot support several young

▣ Monogamous

- One male mates with one female
- Occurs when males have limited mating opportunities, territoriality exists, and the male is certain the offspring are his



□ Sexual Selection

- ▣ Adaptive changes in females and males that lead to differential reproductive success
- ▣ Sexual selection often results in
 - Female choice
 - Male competition



Behaviors that Increase Fitness

□ Societies and Fitness

- Living in a society has a greater reproductive benefit than reproductive cost
 - Benefits of group living include avoiding predators, rearing offspring, and finding food more easily
 - Group living can result in disputes over feeding places and sleeping sites
- Dominance hierarchies are a way to apportion resources
 - Higher-ranking individuals have greater access to essential resources
 - Males and females may form separate dominance hierarchies

Behaviors that Increase Fitness

- **Altruism Versus Self-Interest**
- **Altruism**
 - ▣ Behavior that involves a reduction in individual fitness
 - ▣ Loss may be compensated by an increase in the fitness of another member of the society
- **Inclusive Fitness**
 - ▣ Reproductive fitness of self, and
 - ▣ Reproductive fitness of relatives
 - Meerkat watching over “nieces and nephews” – increased fitness for family genetic line
- Genetic relatedness may underlie many/most acts of apparent altruism
- **Reciprocal altruism** occurs in groups of animals that are mutually dependent
 - ▣ Bird staying behind in nest to help parent raise siblings from a more recent clutch
 - Increased family fitness - genes preserved
 - Chance at acquisition of territory by helping

